

TUNNEL STRUCTURAL HEALTH MONITORING

LiDAR Point Cloud Analysis Report

Generated: 2026-06-10 16:39 | Software: Tunnel Analysis v4.0 - CBNU Smart Structure Lab

Project:	C:/Users/ssl/Desktop/Code Python/data python cursor/tunnel_project/data/full_test/Tn_full.txt
Location:	Osong Test Line
Engineer:	CBNU Smart Structure Lab
Date:	2026-06-10
Profile:	Circle
Scan file:	Tn_full.txt
Points:	150,740
Sections:	80

CBNU Smart Structure Lab

Chungbuk National University · Structural Health Monitoring

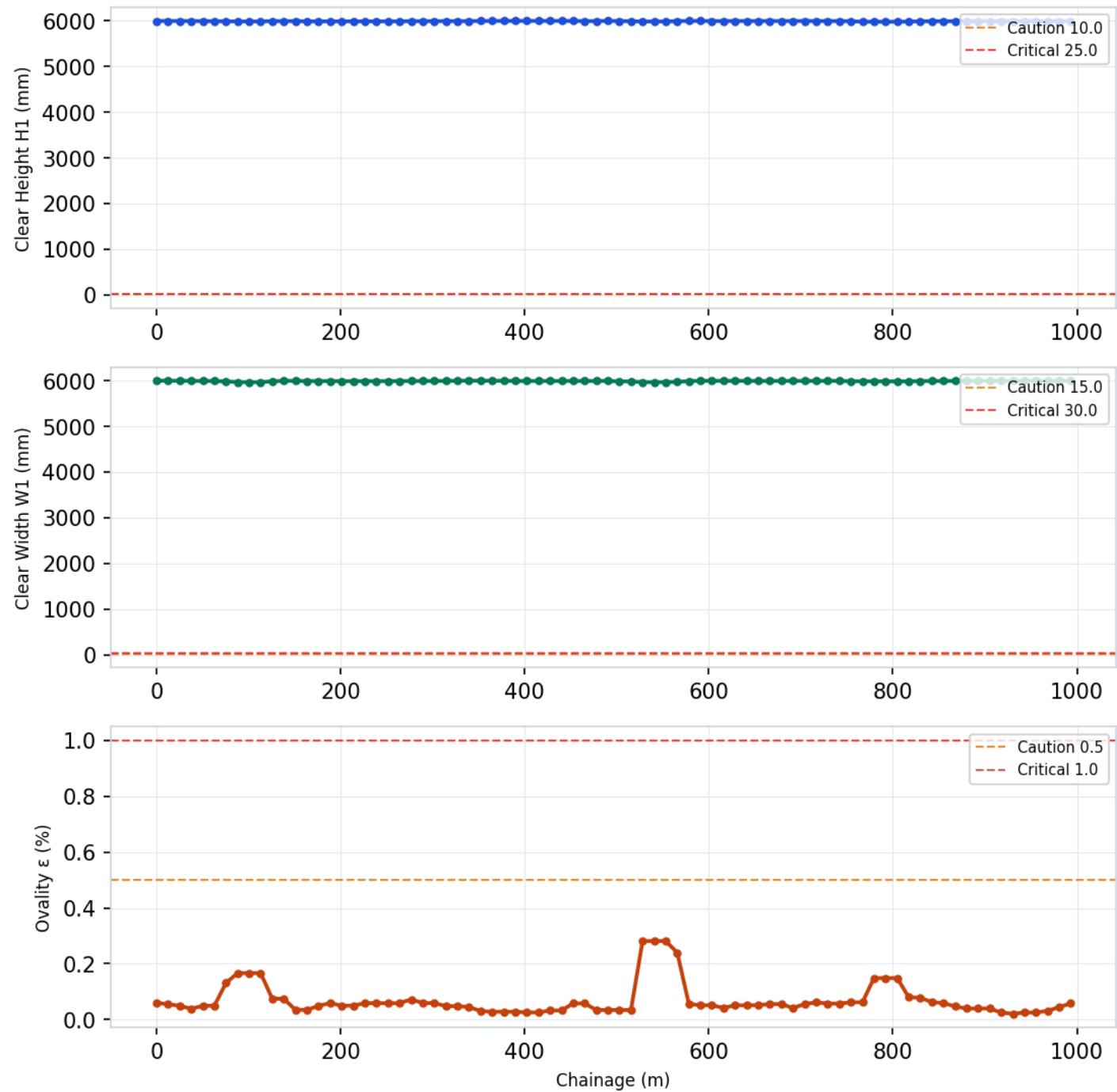
Global Parameters

Parameter	Value	Unit	Threshold	Status
Crown Settlement δv	2.988	mm	C:10.0 / R:25.0	OK
Horizontal Convergence	5.716	mm	C:15.0 / R:30.0	OK
Ovality ϵ (mean)	nan	%	C:0.5 / R:1.0	N/A
Eccentricity e (mean)	7.878	mm	C:10.0 / R:25.0	OK

Section Statistics

Parameter	Min	Max	Mean	Unit	Status
Clear Height H1	5.980	5.999	5.993	m	-
Clear Width W1	5.966	5.998	5.989	m	-
Ovality ϵ	0.021	0.281	0.069	%	-
Eccentricity e	2.312	324.591	15.892	mm	-
Fitted Radius R	2.992	3.007	2.999	m	-

Deformation Trend Along Tunnel Axis



Ch(m)	H1(m)	W1(m)	Oval(%)	Ecc(mm)	R_fit(m)	Clr(m)	Status
0.00	5.997	5.998	0.06	298.7	3.004	-0.839	VIOL
12.57	5.997	5.997	0.06	151.4	3.002	-0.617	VIOL
25.13	5.997	5.996	0.05	4.0	3.000	-0.423	VIOL
37.70	5.996	5.993	0.04	4.0	3.000	-0.223	VIOL
50.27	5.993	5.992	0.05	3.5	3.000	-0.040	VIOL
62.85	5.993	5.991	0.05	3.5	2.999	0.131	OK
75.42	5.990	5.982	0.13	4.0	2.997	0.290	OK
87.99	5.987	5.966	0.17	4.4	2.992	0.410	OK
100.57	5.987	5.966	0.17	4.4	2.992	0.539	OK
113.14	5.987	5.966	0.17	4.7	2.992	0.694	OK
125.72	5.994	5.984	0.07	4.7	2.999	0.822	OK
138.29	5.994	5.995	0.07	4.7	3.000	0.931	OK
150.87	5.994	5.995	0.04	6.5	3.000	1.022	OK
163.45	5.989	5.990	0.04	8.2	3.000	1.110	OK
176.02	5.989	5.990	0.05	8.2	3.000	1.186	OK
188.60	5.989	5.990	0.06	8.2	3.000	1.255	OK
201.18	5.989	5.988	0.05	6.5	3.000	1.316	OK
213.75	5.989	5.988	0.05	5.6	3.000	1.368	OK
226.33	5.989	5.989	0.06	4.1	3.000	1.376	OK
238.91	5.989	5.988	0.06	5.6	3.000	1.339	OK
251.48	5.989	5.989	0.06	5.6	3.000	1.305	OK
264.06	5.989	5.990	0.06	5.6	3.000	1.279	OK
276.63	5.994	5.991	0.07	6.8	3.000	0.261	OK
289.21	5.994	5.991	0.06	6.8	3.000	1.240	OK
301.78	5.994	5.993	0.06	2.7	3.000	1.227	OK
314.36	5.994	5.993	0.05	2.7	3.000	1.223	OK
326.93	5.994	5.996	0.05	2.7	3.000	1.217	OK
339.51	5.994	5.998	0.05	2.7	3.000	1.216	OK
352.08	5.998	5.998	0.03	4.3	3.000	1.219	OK
364.65	5.998	5.996	0.03	4.3	3.000	1.222	OK
377.22	5.999	5.996	0.03	4.3	3.000	1.226	OK
389.80	5.999	5.991	0.03	4.3	3.000	1.235	OK
402.37	5.999	5.991	0.03	2.4	3.000	1.244	OK
414.94	5.999	5.991	0.03	2.3	3.000	1.254	OK
427.51	5.999	5.992	0.03	2.3	3.000	1.263	OK
440.08	5.998	5.992	0.03	2.3	3.000	1.273	OK
452.65	5.998	5.992	0.06	3.6	3.000	1.282	OK
465.22	5.993	5.992	0.06	3.9	3.000	1.289	OK
477.79	5.996	5.992	0.03	3.9	3.000	1.295	OK
490.35	5.998	5.993	0.03	3.9	3.000	1.296	OK
502.92	5.996	5.983	0.03	3.9	3.000	1.297	OK
515.49	5.991	5.983	0.03	3.9	2.999	1.293	OK
528.06	5.986	5.966	0.28	4.7	2.995	1.284	OK
540.63	5.986	5.966	0.28	4.7	2.995	1.244	OK
553.20	5.986	5.966	0.28	4.7	2.995	1.235	OK
565.77	5.995	5.972	0.24	4.7	2.995	1.258	OK
578.34	5.998	5.990	0.06	2.9	3.000	1.257	OK

Ch(m)	H1(m)	W1(m)	Oval(%)	Ecc(mm)	R_fit(m)	Clr(m)	Status
590.91	5.998	5.995	0.05	2.4	3.000	1.247	OK
603.48	5.995	5.995	0.05	2.9	3.000	1.240	OK
616.05	5.995	5.994	0.04	2.9	3.000	1.232	OK
628.63	5.997	5.994	0.05	2.4	3.000	1.226	OK
641.20	5.996	5.994	0.05	2.5	3.000	1.223	OK
653.77	5.996	5.992	0.05	2.5	3.000	1.218	OK
666.35	5.996	5.992	0.06	2.5	3.000	1.218	OK
678.92	5.996	5.992	0.06	2.4	3.000	1.221	OK
691.49	5.996	5.994	0.04	2.4	3.000	1.229	OK
704.07	5.996	5.994	0.06	2.4	3.000	1.242	OK
716.64	5.996	5.994	0.06	3.1	3.000	1.255	OK
729.22	5.996	5.994	0.06	3.1	3.000	1.276	OK
741.80	5.994	5.994	0.06	3.4	3.000	1.306	OK
754.37	5.992	5.987	0.06	3.4	3.000	1.337	OK
766.95	5.980	5.986	0.06	5.1	2.999	1.373	OK
779.53	5.980	5.986	0.15	5.1	2.995	1.368	OK
792.10	5.980	5.985	0.15	6.7	2.995	1.307	OK
804.68	5.980	5.985	0.15	7.5	2.995	1.247	OK
817.26	5.984	5.985	0.08	12.4	2.998	1.190	OK
829.83	5.991	5.990	0.08	12.4	3.000	1.112	OK
842.41	5.992	5.992	0.06	6.7	3.000	1.024	OK
854.99	5.992	5.992	0.06	5.4	3.000	0.930	OK
867.56	5.992	5.993	0.05	4.9	3.000	0.826	OK
880.14	5.993	5.993	0.04	3.1	3.000	0.705	OK
892.71	5.993	5.994	0.04	4.2	3.000	0.578	OK
905.29	5.994	5.994	0.04	4.2	3.000	0.439	OK
917.86	5.994	5.994	0.03	2.9	3.000	0.292	OK
930.43	5.994	5.994	0.02	2.9	3.000	0.127	OK
943.00	5.993	5.993	0.03	3.0	3.000	-0.047	VIOL
955.57	5.993	5.993	0.03	3.0	3.000	-0.230	VIOL
968.14	5.993	5.993	0.03	3.3	3.000	-0.435	VIOL
980.71	5.991	5.994	0.04	163.9	3.003	-0.629	VIOL
993.28	5.993	5.996	0.06	324.6	3.007	-0.856	VIOL

Chainage	Parameter	Value	Status	Action
0.00m	crown_settlement_mm	5997.006	CRITICAL	Immediate inspection
0.00m	lateral_convergence_mm	5998.373	CRITICAL	Immediate inspection
0.00m	eccentricity_mean_mm	298.690	CRITICAL	Immediate inspection
0.00m	clearance_violation	-0.839	CRITICAL	Immediate inspection
12.57m	crown_settlement_mm	5997.128	CRITICAL	Immediate inspection
12.57m	lateral_convergence_mm	5996.977	CRITICAL	Immediate inspection
12.57m	eccentricity_mean_mm	151.358	CRITICAL	Immediate inspection
12.57m	clearance_violation	-0.617	CRITICAL	Immediate inspection
25.13m	crown_settlement_mm	5997.006	CRITICAL	Immediate inspection
25.13m	lateral_convergence_mm	5995.581	CRITICAL	Immediate inspection
25.13m	clearance_violation	-0.423	CRITICAL	Immediate inspection
37.70m	crown_settlement_mm	5996.174	CRITICAL	Immediate inspection
37.70m	lateral_convergence_mm	5993.263	CRITICAL	Immediate inspection
37.70m	clearance_violation	-0.223	CRITICAL	Immediate inspection
50.27m	crown_settlement_mm	5993.142	CRITICAL	Immediate inspection
50.27m	lateral_convergence_mm	5992.050	CRITICAL	Immediate inspection
50.27m	clearance_violation	-0.040	CRITICAL	Immediate inspection
62.85m	crown_settlement_mm	5993.142	CRITICAL	Immediate inspection
62.85m	lateral_convergence_mm	5991.229	CRITICAL	Immediate inspection
75.42m	crown_settlement_mm	5989.889	CRITICAL	Immediate inspection
75.42m	lateral_convergence_mm	5982.127	CRITICAL	Immediate inspection
87.99m	crown_settlement_mm	5986.934	CRITICAL	Immediate inspection
87.99m	lateral_convergence_mm	5966.179	CRITICAL	Immediate inspection
100.57m	crown_settlement_mm	5986.934	CRITICAL	Immediate inspection
100.57m	lateral_convergence_mm	5966.179	CRITICAL	Immediate inspection
113.14m	crown_settlement_mm	5986.934	CRITICAL	Immediate inspection
113.14m	lateral_convergence_mm	5966.179	CRITICAL	Immediate inspection
125.72m	crown_settlement_mm	5993.925	CRITICAL	Immediate inspection
125.72m	lateral_convergence_mm	5984.014	CRITICAL	Immediate inspection
138.29m	crown_settlement_mm	5993.925	CRITICAL	Immediate inspection